Task 3: Exploratory Data Analysis

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1. Introduction

The purpose of this task is to perform exploratory data analysis on the cleaned e-commerce dataset.

Exploratory Data Analysis, also known as EDA, is used to understand patterns, trends, and important relationships in a dataset. It helps identify useful insights before creating visualizations or building prediction models.

The dataset used for this task contains customer order information, including products, quantities, prices, payment methods, order statuses, coupon codes, referral sources, and total prices.

2. Dataset Overview

The dataset contains:

Item	Value
Number of rows	1200
Number of columns	14
Missing Values	0
Duplicate Rows	0
Date Range	2023-01-01 to 2025-06-30

Table 1: Dataset Overview Table

Since the missing values in the CouponCode column were already replaced with "No Coupon" in task 2, the dataset is ready for analysis.

3. Basic Numerical Summary

The main numerical columns in the dataset are Quantity, UnitPrice, ItemsInCart, and TotalPrice.

Column	Average	Minimum	Maximum
Quantity	2.95	1	5
UnitPrice	356.41	11.39	699.93
ItemsInCart	5.48	1	10
TotalPrice	1053.97	11.39	3456.40

Table 2: Numerical Summary Table

The average order value is approximately 1053.97. The minimum total price is 11.39, while the maximum total price is 3456.40.

This shows that order values vary widely depending on the product price and quantity purchased.

4. Product Analysis

The dataset includes seven product categories.

Product	Number of Orders
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

Table 3: Product Analysis Table

The most frequently ordered product is **Printer**, followed closely by **Tablet** and **Chair**. The least frequently ordered product is **Phone**.

5. Revenue by Product

The total revenue generated by each product was also analysed.

Product	Total Revenue
Chair	195620.11
Printer	195612.61
Laptop	192126.56
Tablet	186568.95
Monitor	175651.41
Desk	167459.93
Phone	151722.39

Table 4: Total Revenue by Product Table

The product that generated the highest total revenue is the Chair. Following very closely, is the Printer. Even though the printer had the highest number of orders, the chair generated slightly more total revenue. On the other hand, phone generated the lowest revenue among the listed products.

6. Order Status Analysis

The order statuses in the dataset are:

Order Status	Number of Orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Table 5: Order Status Table

The most common order status is **Cancelled**, followed by **Returned**. This may be an important issue for the business because a high number of cancelled and returned orders can reduce successful sales.

7. Payment Method Analysis

The dataset includes five payment methods.

Payment Method	Number of Orders
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Table 6: Payment Method Analysis Table

The most used payment method is **Online**, followed by **Cash**. The difference between payment methods is not very large, meaning customers use a variety of payment options.

8. Coupon Code Analysis

The CouponCode column was cleaned by replacing missing values with "No Coupon".

Coupon Code	Number of Orders
FREESHIP	313
No Coupon	309
WINTER15	292
SAVE10	286

Table 7: Coupon Code Analysis Table

The most used coupon code is **FREESHIP**. Many orders also had **No Coupon**, which means many customers placed orders without using any discount code.

This shows that coupons may influence customer purchases, but they are not used in every order.

9. Referral Source Analysis

The dataset shows where customers came from before placing an order.

Referral Source	Number of Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Table 8: Referral Source Analysis Table

The most common referral source is **Instagram**, followed by **Email** and **Google**. This suggests that social media and email marketing may be important sources of customer traffic.

10. Relationship Between Numerical Variables

The relationship between numerical columns was reviewed.

Important observations:

Relationship	Result
UnitPrice and TotalPrice	Strong positive relationship
Quantity and TotalPrice	Positive relationship
Quantity and ItemsInCart	Positive relationship
ItemsInCart and TotalPrice	Moderate positive relationship

Table 9: Relationship Between Numerical Variables Table

This makes sense because TotalPrice is affected by both UnitPrice and Quantity. Higher unit prices and larger quantities usually lead to higher total order of prices.

11. Outlier Check

The TotalPrice column was checked for unusually high or low values.

Most total prices are within a normal range, but a small number of high-value orders can be considered outliers. These outliers are not necessarily errors because they may simply represent expensive products or larger quantities.

For this reason, the outliers should be kept in the dataset unless there is clear evidence that they are incorrect.

12. Key Findings

The main findings from the exploratory data analysis are:

Area	**Finding**
Product orders	Printer has the highest number of orders
Product revenue	Chair generated the highest total revenue
Order status	Cancelled is the most common order status
Payment method	Online is the most used payment method
Coupon usage	FREESHIP is the most used coupon code
Referral source	Instagram is the top referral source
Total price	TotalPrice increases with Quantity and UnitPrice

Table 10: Summary Table

13. Task 3 Summary

In this task, exploratory data analysis was performed on the cleaned e-commerce dataset.

The dataset contains 1200 orders and 14 columns. The analysis showed that Printer had the highest number of orders, while Chair generated the highest total revenue. Online payment was the most common payment method, and Instagram was the leading referral source.

The CouponCode column showed that FREESHIP was the most used coupon, while many customers also placed orders without using any coupon. The order status analysis showed that Cancelled and Returned orders were common, which may be an important business concern.

Overall, this exploratory analysis helped identify important patterns in product sales, revenue, customer behavior, payment methods, coupon usage, and order status.